Visualizing Multivariate Time Series Data in an Environmental Input/Output Systems

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Application, Users, and Tasks

Microalgae bioreactors (Figure 1) improve indoor air quality by sequestering carbon, measured as the difference between CO_2 input and output. Their effectiveness depends on solar exposure and other environmental factors. **Sustainable architects and building managers** seek a visualization system to: (1) Optimize performance by analyzing carbon reduction time series in relation to environmental inputs (ie. light, CO_2, etc.), and (2) Engage the public by showcasing how the system reduces indoor air pollution through sequestration of CO_2.

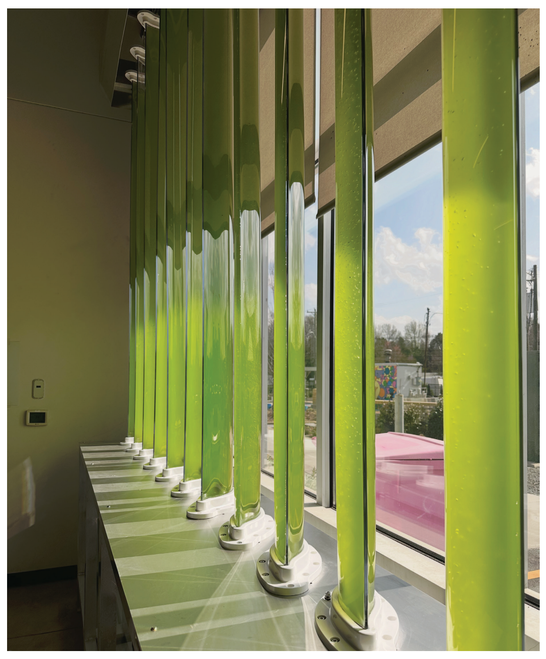


Figure 1: A row of bioreactors growing microalgae facing southwest

Requirements

- **Correlation Explored Over Time**: Effectively show relationships between carbon reduction and environmental factors over diurnal cycles and across multiple days.
- **Subtle Change Detection**: Reveal small changes in CO_2 input/output differences.
- **Scalability**: Ensure an uncluttered display that can scale to a large number of days.
- **Public Friendliness:** Easy to understand, and accessible by the general public.

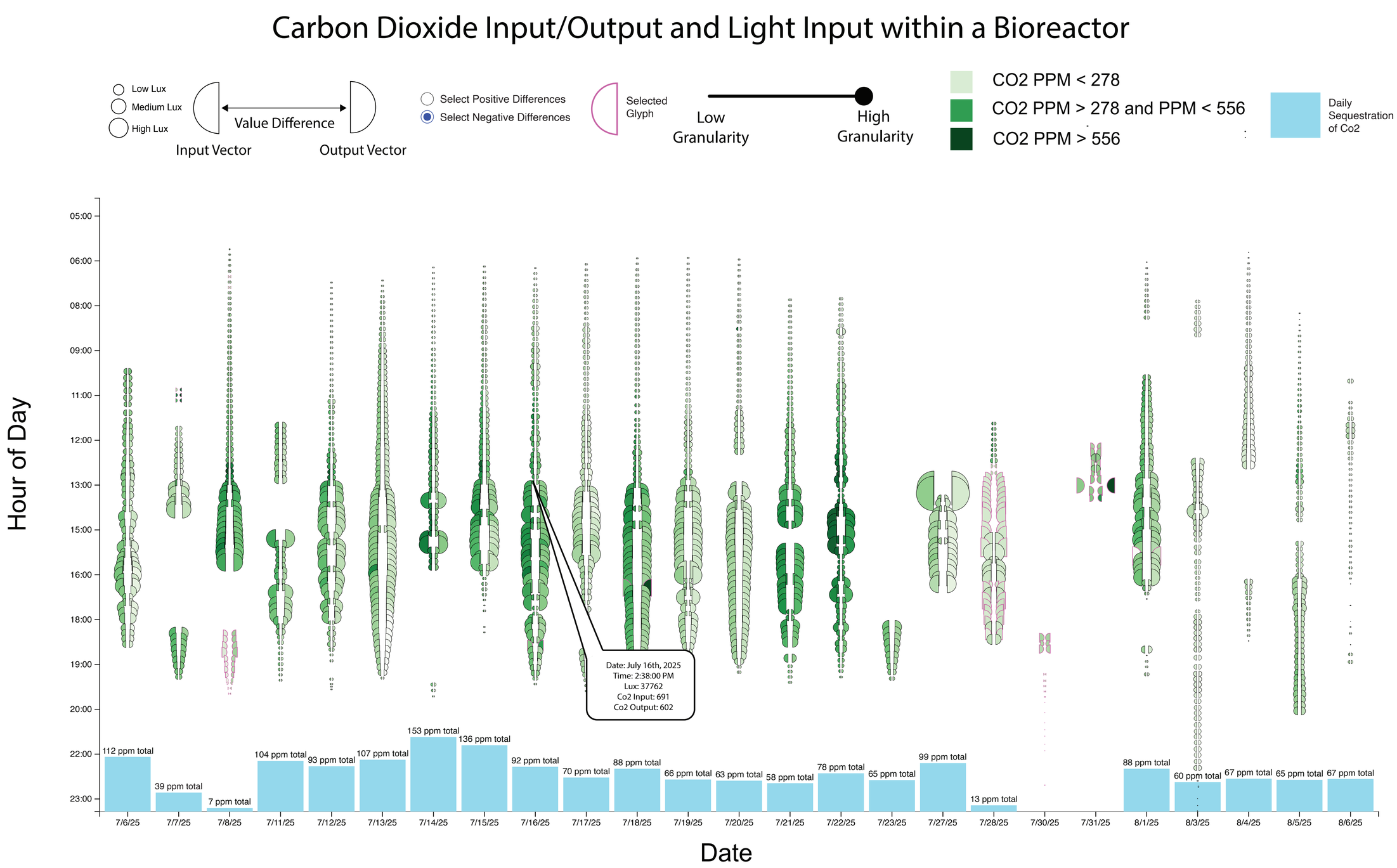


Figure 2: CO_2 input/output data and light data throughout July 2025 from bioreactors. The size of the semicircles encodes light values, while the color indicates concentration of CO_2 The distance between the two semicircles represents the difference in CO_2 input and output, with negative differences highlighted in magenta. The height of the bar at the bottom encodes daily aggregated CO2 sequestration. The figure reveals the bioreactor sequesters the most CO2 in the afternoons when sunlight is most intense.

Design

- **Glyph, Color, and Size:** A single point is split into an input and output glyph (semicircle), with color denoting CO_2 concentration, size indicating lux value, and semicircle orientation representing input or output.
- **Difference between CO_2 in and out :** $(Co2.in - Co2.out)$ is encoded by the distance between the input and output semicircles, scaled with the inverse hyperbolic sine function $\operatorname{arcsinh} x$ to reduce clutter. Positive and negative differences are distinguished by the orientation of the semicircles.
- **Interaction:** Users can select the time granularity and highlight positive or negative differences.

Expert Interviews

Feedback from a sustainable architect and a building manager:

Positive

- Graph helps observe correlation quickly.
- Distance between semicricles and colors intuitively reveal the CO_2 difference.
- Aggregated CO_2 bar instantly reveals which days the most $CO_2$was sequestered.

Constructive

- Glyphs were confusing at first. A clearer way of introducing them would be nice.

Conclusion

- We propose a novel visual metaphor to visualize the difference between inputs and outputs of natural systems over time using semicircles and the distance between them as methods of visual encoding.
- We built a working time series visualization prototype to explore the correlation between carbon reduction and environmental factors.

Future Work

- Explore different ways to enhance scalability (e.g. aggregation, filtering, etc).
- Explore alternative visual encoding for positive and negative differences.
- Further incorporate interaction design to aid user tasks such as causal inference.

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